IGCSE 0607 Practice Questions — E1.6

The four operations

Total Questions: 18 — **Total Marks:** 99 Marks

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1 Work out the exact value of

$$-3 - 4 \times \left(\frac{1}{2} - \frac{3}{4}\right)^2$$

..... **[3]**

2 **(a)** The temperature at midnight was $-5°C$.
By noon, the temperature had risen by $14°C$.
Write down the temperature at noon.

..... **[1]**

(b) Work out the difference between $-8°C$ and $17°C$.

..... **[1]**

(c) Work out the value of

$$24 - 12 \div (3 - 7)$$

..... **[2]**

3 **(a)** Work out the value of

$$12 - 2 \times (-3) + 4^2$$

..... **[2]**

(b) Work out the value of

$$\frac{3}{4} + \frac{1}{6} \times \frac{3}{2}$$

Give your answer in its simplest form.

..... **[2]**

4 **(a)** Work out the value of

$$12 - 3 \times (-4) + (15 \div -3)$$

..... **[2]**

(b) Work out the value of

$$\frac{2}{5} \div \frac{4}{7} - \frac{1}{10}$$

Give your answer as a fraction in its simplest form.

..... **[2]**

5 **(a)** Work out the value of

$$15 - 3 \times (4 - (-2))$$

..... **[2]**

(b) Work out the value of

$$2\frac{1}{4} \div \frac{3}{8} - 1\frac{1}{2}$$

Give your answer in its simplest form.

..... **[3]**

6 **(a)** Work out the value of

$12 - 4 \times (5 - 9)$

..... **[2]**

(b) Work out the value of

$1.2 \div 0.04$

..... **[2]**

(c) The temperature at 04:00 was $-5.2°C$.
By 14:00, the temperature had risen by $8.7°C$.
Work out the temperature at 14:00.

..... °C **[1]**

7 **(a)** Work out the exact value of

$\frac{5}{6} + \frac{2}{3} \times \frac{9}{10}$ Give your answer as a mixed number in its simplest form.

..... **[3]**

(b) Work out the exact value of

$2\frac{1}{4} \div \left(1\frac{1}{8} - \frac{3}{4}\right)$

..... **[3]**

8 **(a)** Calculate the exact value of

$$2\frac{2}{5} \div \left(1\frac{1}{3} - \frac{5}{6}\right)$$

Show all your working and give your answer as a mixed number in its simplest form.

..... **[4]**

(b) Work out the exact value of

$$\frac{0.45 \times 1.2}{0.06}$$

..... **[3]**

9 Solve the equation, giving your answer as a fraction in its simplest form.

$$\frac{x+\frac{1}{3}}{2-0.4} = -1\frac{1}{4}$$

$x =$ ..... **[5]**

10 Work out the exact value of

$$\frac{2\frac{1}{3} \times 1\frac{2}{7} - \sqrt{\frac{25}{9}}}{\left(\frac{1}{2}\right)^3 \div \frac{3}{16}}$$

..... **[5]**

11 Without using a calculator, work out

$$\frac{3\frac{2}{3} + 2\frac{1}{4} \times 1\frac{1}{3}}{4\frac{1}{2} - 1\frac{5}{6}}$$

You must show all your working and give your answer as a fraction in its simplest form.

..... **[5]**

12 **(a)** Work out the value of

$$2\frac{2}{3} \div \left(1\frac{1}{4} - \frac{5}{6}\right)$$

Give your answer as a mixed number in its simplest form.

..... **[4]**

(b) A rectangular field has a length of $12\frac{1}{2}$ m and an area of $83\frac{1}{3}$ m^2.
Work out the perimeter of the field.
Give your answer as a mixed number in its simplest form.

..... m **[4]**

13 **(a)** At 06:00, the temperature in a city is $-4.5°C$.
By 14:00, the temperature has risen by $11.2°C$.
By 22:00, the temperature has dropped by $8.1°C$ from the temperature at 14:00.

Work out the temperature at 22:00.

..... **[2]**

(b) Use your calculator to work out the value of

$$\frac{4.72^2 - \sqrt[3]{18.5}}{3.1 \times (0.45 + 1.2)}$$

Give your answer correct to 3 significant figures.

..... **[2]**

14 A water tank has a capacity of 85.5 litres.
It is initially completely empty.
A hose pumps water into the tank at a rate of $\frac{3}{4}$ of a litre per minute.
At the same time, a leak in the tank loses water at a rate of 0.45 litres per minute.

Work out how long it takes to completely fill the tank.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[4]**

15 The temperature of a liquid is $-4.5°C$.
It is heated so that its temperature rises by $2.8°C$ every minute for 15 minutes.

(a) Work out the temperature of the liquid after the first 15 minutes.

..... **[2]**

(b) After the 15 minutes, the liquid is cooled so its temperature falls by $1.2°C$ every minute for m minutes.
After the m minutes of cooling, the temperature of the liquid is $9.9°C$.
Work out the value of m.

$m =$ ..... **[3]**

16 A scientific freezer is used to change the temperature of liquid samples.

(a) Sample A has an initial temperature of 12.5°C.
It cools at a constant rate of 3.2°C per minute.
Calculate the temperature of Sample A after 8 minutes.

..... **[2]**

(b) Sample B has an initial temperature of −4.5°C.
To find its final temperature, the initial temperature is multiplied by −2, and the result is then decreased by 18.3°C.
Calculate the positive difference between the final temperature of Sample B and the final temperature of Sample A.

..... **[4]**

17 A large container has a total capacity of 1000 litres.

(a) The container is initially $\frac{2}{5}$ full of water.
Water is pumped into the container at a constant rate of 18.5 litres per minute for 20 minutes.
Work out the volume of water in the container after these 20 minutes.

..... litres **[3]**

(b) Water then leaks from the container at a constant rate of 4.5 litres per minute.
Calculate the time taken for the container to become exactly half full.

..... minutes **[3]**

(c) Once the container is half full, the leak is fixed.
A different pump then fills the container completely in 1 hour and 20 minutes.
Calculate the average rate at which this pump fills the container.
Give your answer in litres per minute.

..... litres per minute **[3]**

18 **(a)** A tank contains 450.5 litres of oil.
The oil is poured into bottles.
Each bottle holds exactly 1.25 litres.
Work out the number of completely filled bottles and the volume of oil left over.

Number of bottles = ... , Volume left over = ... **[3]**

(b) A machine pumps water at a rate of 1800 litres per minute.
It is used to fill a rectangular pool of length 24.5 m, width 12 m and depth 1.8 m.
($1\ m^3 = 1000$ litres)

Calculate the time taken to fill the pool completely.
Give your answer in hours and minutes.

hours, minutes **[4]**

(c) A freezer is set at $-18°C$.
During a power cut, the temperature inside the freezer rises by $1.5°C$ per hour.
The power cut lasts for 6 hours and 20 minutes.
Work out the temperature in the freezer when the power is restored.

..... °C **[3]**